

Special Relativity Notes

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Contents

1	Preliminaries	4
1.1	Observing	4
1.2	Spacetime Diagrams	4
1.3	Reference Frames	5
1.3.1	Example 1	5
1.3.2	A 's Point of View	5
1.3.3	B 's Point of View	5
1.3.4	Example 2	5
1.3.5	Earth's Point of View	5
1.3.6	A 's Point of View	6
1.3.7	B 's Point of View	6
1.3.8	Inertial Reference Frames	6
1.4	Galilean Transformations	7
2	The Axioms	7
3	Relativity of Simultaneity	7
3.1	Case 1	7
3.2	Case 2	7
3.3	Case 3	8
3.4	Leading Clocks Lag	8
4	Lorentz Factor	8
5	Time Dialation	8
5.1	Light Clocks	8
5.2	Time Dilation	9
5.2.1	A 's Reference Frame	9
5.2.2	B 's Reference Frame	9
5.2.3	Putting it Together	10
6	Length Contraction	11

7	Invariants	11
7.1	Other Lengths	11
7.2	Spacetime Interval	11
7.2.1	A 's Reference Frame	11
7.2.2	B 's Reference Frame	12
7.2.3	D 's Reference Frame	12
7.2.4	Putting it Together	13
8	Common Units for c	13
9	Lorentz Transformations	13
9.1	Time Dilation and Length Contraction Example (Part 1)	13
9.1.1	Earth's Reference Frame	13
9.1.2	Rocket's Reference Frame	13
9.2	Lorentz Transformations (Special Case)	13
9.3	Lorentz Transformations (General Case)	14
9.4	Leading Clocks Lag Revisited	15
9.4.1	Rearward Light Beam	15
9.4.2	Frontward Light Beam	16
9.4.3	Putting it Together	16
9.4.4	Proof with Lorentz Transformation	16
9.5	Time Dilation and Length Contraction Example (Part 2)	17
10	Combining Velocities	17
10.1	Parallel Velocities	17
10.2	Perpendicular Velocities	18
11	Spacetime Diagrams	18
11.1	Transformations	19
11.2	Intervals	20
12	Faster than Light Travel	21
12.1	A Possible Way	21
12.2	Breaking of Causality	21
13	Paradoxes	22
13.1	Pole in the Barn Paradox	22
13.1.1	The Barn's Reference Frame	23
13.1.2	The Pole's Reference Frame	23
13.2	Spaceships on a Rope Paradox	23
13.2.1	The Rockets' Reference Frame	23
13.2.2	A 's Reference Frame	23
13.3	How Does Length Contraction Happen?	23
13.4	The Twin Paradox	24
13.4.1	B 's Reference Frame	24
13.4.2	A 's Reference Frame	24

13.4.3 Paradox and Resolution	25
13.5 General Solutions	25
13.5.1 Pole in the Barn	25
13.5.2 Twin Paradox	25

1 Preliminaries

1.1 Observing

We will measure time with clocks.

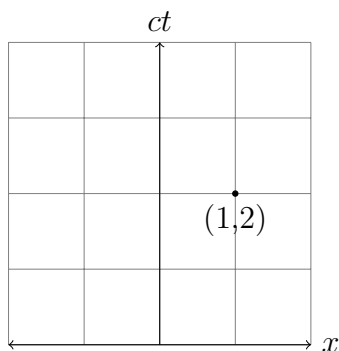
Definition 1.1. The “Photo Clock Principle” is when an event occurs, we assume that there is a “photo” taken that records the position and the time. (Imagine that there are clocks everywhere.)

Definition 1.2. We define Observing an event to be recording that event’s position, and the time that it occurred.

We assume that clocks belonging to person A are in sync relative to person A .

1.2 Spacetime Diagrams

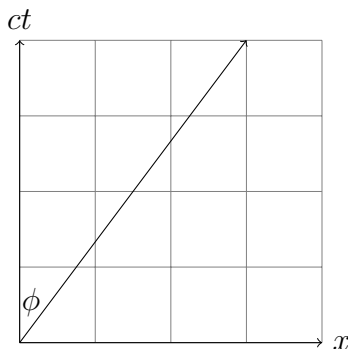
A Spacetime Diagram is a diagram as below.



We represent events as points. Note that time is on the vertical axis.

Definition 1.3. The *worldline* of an object is the path that it makes through spacetime.

Example 1.4. The worldline of an object moving to the right at a constant velocity is below.



The speed of the object is $\tan(\phi) = \frac{1}{s}$ where s is the slope of the line.

If an object is not moving relative to the observer, it has a vertical worldline.

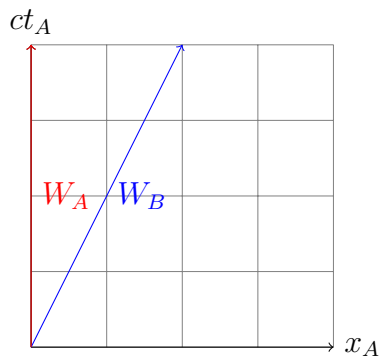
1.3 Reference Frames

1.3.1 Example 1

Let us suppose that A and B are moving relative to each other, and cross each other.

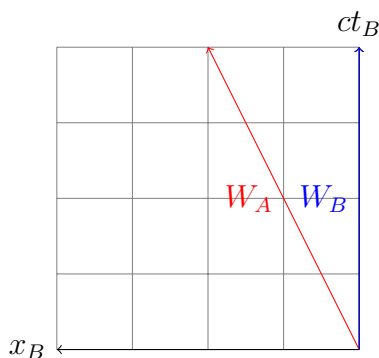
1.3.2 A 's Point of View

From A 's point of view, B flies by her. In her reference frame, the spacetime diagram is below:



1.3.3 B 's Point of View

From B 's Point of view, A flies by him, the spacetime diagram is below:

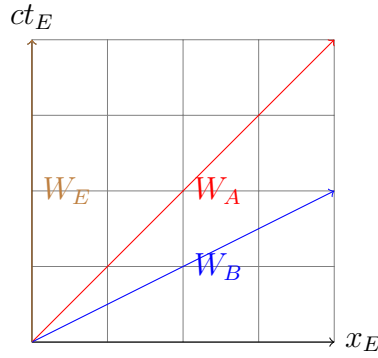


1.3.4 Example 2

Suppose that A and B fly by the Earth such that $v_B = 2v_A$ relative to the Earth.

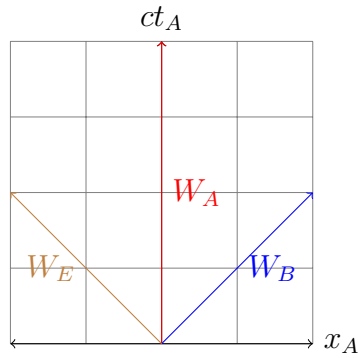
1.3.5 Earth's Point of View

The Earth sees A flying away from it, and B flying faster. The spacetime diagram is below:



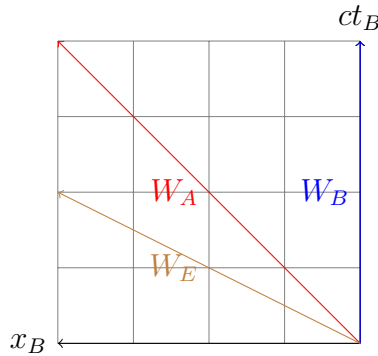
1.3.6 A's Point of View

From A 's point of view, B is flying away from her ahead of her, and the Earth is retreating behind her. The spacetime diagram is below:



1.3.7 B's Point of View

From B 's point of view, both A and the Earth are retreating behind him, with the Earth twice as fast as A . The spacetime diagram is below:



1.3.8 Inertial Reference Frames

Definition 1.5. An Inertial Reference Frame is a reference frame that is either at rest or moving with a constant velocity. Special Relativity only deals with Inertial Reference Frames.

1.4 Galilean Transformations

Given an event in B 's RF, we want to find it in A 's RF. Let v_B be B 's velocity relative to A . Assume that, when $t_A = t_B = 0$, $x_A = x_B = 0$. Then, $(x_A)_{event} = (x_B)_{event} + v_B \cdot t_B$, and $t_A = t_B$.

2 The Axioms

Axiom 1 (Principle of Relativity). All observers moving at a constant velocity can claim to be stationary.

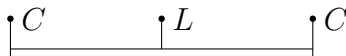
Axiom 2 (Principle of Light Constancy). The speed of light is not observer-dependant.

Alternate phrasings of Axiom 1 are that it is impossible to conduct an experiment to determine if you are moving at a constant velocity, and that there is no state of absolute rest. An alternate phrasing of Axiom 2 is that the speed of light is the same for all observers.

Notation. We write c for the speed of light, which is about $3 \times 10^8 \frac{m}{s}$ ¹.

3 Relativity of Simultaneity

We will assume that A has a spaceship like the one below.



With clocks at points labeled C , and a launcher at the point labeled L .

3.1 Case 1

Let us suppose that A is stationary relative to B , and that her launcher shoots out light pulses. Assuming that $t_A = t_B = 0$, then A and B will see the light reach the front and the back at the same time.

3.2 Case 2

Let us suppose that A is moving to the right with velocity v relative to B , and that her launcher shoots out paintballs in opposite directions with velocity u . Then, the paintball moving towards the right has velocity $u + v$ (right) relative to B , and the one moving left has velocity $u - v$ (left) relative to B ². However, the rightwards traveling paintball has to travel more distance than the leftwards one, and it turns out that this exactly cancels out.

¹299,792,458

²For Small Velocities

3.3 Case 3

Let us assume the same setup as Case 2, but instead of paintballs, we let the launcher shoot out light pulses. Naturally, A can claim to be stationary, so she sees the pulses reach the ends at the same time. However, to B , the rightward pulse has to travel more than the leftward one, so (as they both have speed c), the leftward one hits the spaceship first. This means that what A observes as simultaneous, B does not.

Theorem 3.1. *Simultaneity is relative.*

Corollary 3.2. *There is no absolute time scale.*

3.4 Leading Clocks Lag

Let us return to Case 3. Let t_L and t_R be the times that B observes the leftward and rightward pulse to reach the end respectively. Then, $t_R > t_L$. However, both times, A 's respective clock reads t_A . This has an important consequence.

Theorem 3.3 (Leading Clocks Lag). *If A is moving relative to B , and A 's clocks are in sync relative to A , then B will observe A 's leading clock (the one in front) to lag behind A 's rear clock.*

4 Lorentz Factor

Before going further, we must define the Lorentz Factor.

Definition 4.1. The Lorentz Factor is

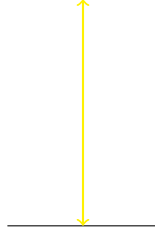
$$\gamma := \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}}.$$

The velocity v will be clear from context.

5 Time Dialation

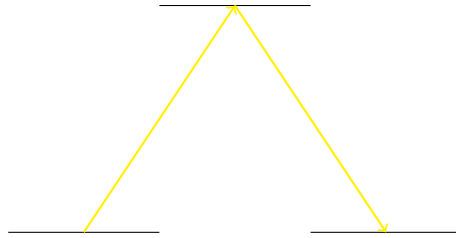
5.1 Light Clocks

Definition 5.1. We define a *light clock* to be a setup like below.



Where the yellow represents a light beam. One tick of the light clock corresponds to the beam completing a cycle (i.e. bouncing up and down).

Now let us suppose that both A and B have light clocks, and that A is moving to the right relative to B . B will then see A 's clock as below.



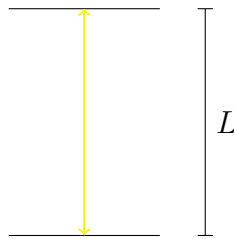
This means that B will see A 's light beam traveling further (with speed c), and will therefore see A 's clock as running slower than his. This is called *Time Dilation*.

5.2 Time Dilation

We want to find the exact discrepancy between t_A and t_B . We assume that A is moving right relative to B with velocity v .

5.2.1 A 's Reference Frame

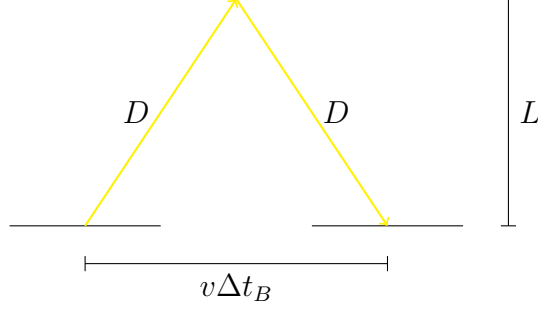
To A , the light clock will be perfectly normal.



She will therefore see a tick as $\Delta t_A = \frac{2L}{c}$. We will use the equivalent equation $L = \frac{c\Delta t_A}{2}$.

5.2.2 B 's Reference Frame

To B , the light clock will be different.



We can see that a tick for B is $\Delta t_B = \frac{2D}{c}$ (or $D = \frac{c\Delta t_B}{2}$). We can also see that

$$D^2 = L^2 + \left(\frac{v\Delta t_B}{2}\right)^2 = L^2 + \frac{1}{4}v^2\Delta t_B^2.$$

5.2.3 Putting it Together

We know that $D^2 = L^2 + \frac{1}{4}v^2\Delta t_B^2$, $D = \frac{c\Delta t_B}{2}$, and that $L = \frac{c\Delta t_A}{2}$. We put these together to get

$$\begin{aligned} \left(\frac{c\Delta t_B}{2}\right)^2 &= \left(\frac{c\Delta t_A}{2}\right)^2 + \frac{1}{4}v^2\Delta t_B^2 \\ \Rightarrow \frac{1}{4}c^2\Delta t_B^2 &= \frac{1}{4}c^2\Delta t_A^2 + \frac{1}{4}v^2\Delta t_B^2 \\ \Rightarrow c^2\Delta t_B^2 &= c^2\Delta t_A^2 + v^2\Delta t_B^2 \\ \Rightarrow \Delta t_B^2(c^2 - v^2) &= c^2\Delta t_A^2 \\ \Rightarrow \Delta t_B^2 &= \frac{c^2}{c^2 - v^2}\Delta t_A^2 = \frac{1}{1 - \frac{v^2}{c^2}}\Delta t_A^2 \\ \Delta t_B &= \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}}\Delta t_A = \gamma\Delta t_A. \end{aligned}$$

This is quite important.

Theorem 5.2 (Time Dilation). *If we write Δt_M for the elapsed time measured by someone moving, and Δt_R for the elapsed time measured by someone at rest, then the following is true:*

$$\Delta t_R = \frac{\Delta t_M}{\sqrt{1 - \frac{v^2}{c^2}}} = \gamma\Delta t_M.$$

Remark 5.3. Let us suppose that M measures 1 unit of time. Person R will then measure γ units. Then, $\gamma \cdot T_R = 1 \cdot T_M$ where T_R is one unit of time for person R (M respectively).

6 Length Contraction

Let us suppose that B flies by A in a ship at velocity v . Let us say that A measures the front of the ship passing her at time t_{A1} , and the end at t_{A2} . B will also measure these (say t_{B1} and t_{B2} respectively). Then, A will measure the length as $L_A = v(t_{A2} - t_{A1}) = v\Delta t_A$, and B will measure $L_B = v(t_{B2} - t_{B1}) = v\Delta t_B$. Although B is moving relative to A , he is stationary relative to the thing that we are measuring, so we will assume B to be at rest. Then, $\Delta t_B = \gamma\Delta t_A$, so $L_B = v\Delta t_B = v\gamma\Delta t_A = \gamma L_A$. This is also quite important.

Theorem 6.1 (Length Contraction). *Let us suppose that we have an object O . Let L_R be the length of O in the x -direction measured by an observer that is not moving relative to O , and let L_M be the length of O in the x -direction measured by an observer moving with velocity v in the x -direction relative to O . Then, $L_R = \gamma L_M$.*

7 Invariants

7.1 Other Lengths

Let assume that our motion is only in the x -direction. We claim that there is then no contraction in the y -direction. Consider the diagram below.



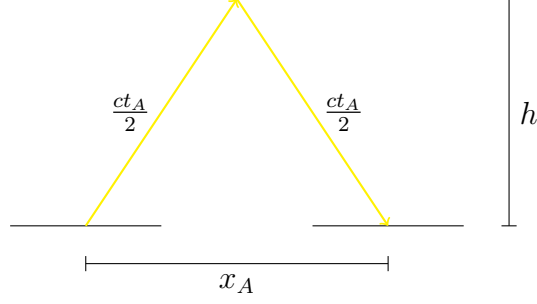
If there was contraction in the y -direction, then, from the Train's Reference Frame, the tracks would become too small for it to move on. However, from the Tracks' Reference Frame, the train becomes too small for them. These are different things, and cannot both happen at the same time. Therefore, there is no contraction in the y -direction. By symmetry, the same is true for z .

7.2 Spacetime Interval

Let us suppose that B is stationary and has a light clock, A is moving to the left with velocity v , and D is moving to the right with velocity u . Let the light leaving the bottom be event E_1 , and it hitting the bottom again be event E_2 . We examine their reference frames.

7.2.1 A 's Reference Frame

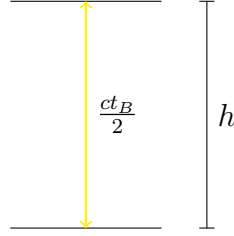
We define x_A to be the distance A measures between E_1 and E_2 , and t_A the time between them according to A .



We can see that $h^2 + \left(\frac{x_A}{2}\right)^2 = \left(\frac{ct_A}{2}\right)^2$, which means that $4h^2 + x_A^2 = c^2t_A^2$, or that $4h^2 = c^2t_A^2 - x_A^2$.

7.2.2 B's Reference Frame

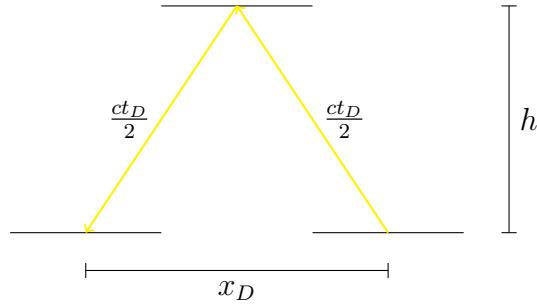
We define x_B to be the distance B measures between E_1 and E_2 , and t_B the time between them according to B .



We can see that $h^2 = \left(\frac{ct_B}{2}\right)^2$, which means that $4h^2 = c^2t_B^2$. Because $x_B = 0$, we have that $4h^2 = c^2t_B^2 - x_B^2$.

7.2.3 D's Reference Frame

We define x_D to be the distance D measures between E_1 and E_2 , and t_D the time between them according to D .



We can see that $h^2 + \left(\frac{x_D}{2}\right)^2 = \left(\frac{ct_D}{2}\right)^2$, which means that $4h^2 + x_D^2 = c^2t_D^2$, or that $4h^2 = c^2t_D^2 - x_D^2$.

7.2.4 Putting it Together

We have seen that, in the three possible cases (moving left, stationary, and moving right), the quantity $c^2t^2 - x^2$ is the same. This is therefore true for any inertial reference frame.

Theorem 7.1. *Given two events E_1 and E_2 , let t be the time between them as measured by any inertial reference frame. Let x be the distance between them, as measured by the same frame. Then, the quantity $c^2t^2 - x^2$ is the same for all inertial reference frames. In other words, $c^2t^2 - x^2$ is invariant.*

8 Common Units for c

It is common to define a lightyear as the distance that light travels in a year, and write $c = 1 \frac{\text{lightyear}}{\text{year}} = 1$.

9 Lorentz Transformations

9.1 Time Dilation and Length Contraction Example (Part 1)

Suppose that the Earth is stationary relative to a star 5 lightyears away. A will travel from the Earth to the star at speed $v = 0.943c$ ($\gamma \approx 3$).

9.1.1 Earth's Reference Frame

The Earth observer will see the travel time as $\frac{5}{0.943c}$ lightyears ≈ 5.3 years. However, when the rocket reaches the star, Earth will read its clocks as saying $\frac{5.3}{\gamma} = 1.77$ years.

9.1.2 Rocket's Reference Frame

The rocket observer would see the star-Earth system moving. They would see the length contract into $\frac{5}{\gamma}$ lightyears = 1.67 lightyears. They will therefore see the time taken be $\frac{1.67}{0.943c}$ lightyears = 1.77 years. However, when they reach, the rocket observer would see Earth's clocks reading $\frac{1.77}{\gamma}$ years = 0.59 years. We will explain this later.

9.2 Lorentz Transformations (Special Case)

Let us suppose that B is moving with velocity v relative to A . We will also assume that, when $t_A = t_B = 0$, $x_A = x_B = 0$. Now, let B observe an event at time t_B , at $x_B = 0$. We want to find the coordinates of the event in A 's Reference Frame. We have the following:

$$x_B = 0$$

$$t_A = \gamma t_B.$$

We also have that $x_A = vt_A$, so $x_A = \gamma vt_B$.

9.3 Lorentz Transformations (General Case)

We now assume that $x_B \neq 0$. The conversion equations will have the following form:

$$t_A = gx_B + ht_B \qquad x_A = mx_B + nt_B.$$

By letting $x_B = 0$, we get that $h = \gamma$, and $n = \gamma v$. Therefore, we have the following:

$$t_A = gx_B + \gamma t_B \qquad x_A = mx_B + \gamma v t_B.$$

Now recall that $c^2 t_A^2 - x_A^2 = c^2 t_B^2 - x_B^2$. We will use this to find g and m . We have the equations below.

$$\begin{aligned} c^2(gx_B + \gamma t_B)^2 - (mx_B + \gamma v t_B)^2 &= c^2 t_B^2 - x_B^2 \\ &= c^2 (g^2 x_B^2 + 2g\gamma x_B t_B + \gamma^2 t_B^2) - (m^2 x_B^2 + 2m\gamma v x_B t_B + \gamma^2 v^2 t_B^2) \\ &= (c^2 \gamma^2 - v^2 \gamma^2) t_B^2 + (2c^2 \gamma g - 2m\gamma v) x_B t_B + (c^2 g^2 - m^2) x_B^2 \end{aligned}$$

This gives us the following equations.

$$c^2 \gamma^2 - v^2 \gamma^2 = c^2 \qquad 2c^2 \gamma g - 2m\gamma v = 0 \qquad c^2 g^2 - m^2 = -1$$

The equations can be used as below.

$$\begin{array}{lll} c^2 \gamma^2 - v^2 \gamma^2 = c^2 & 2c^2 \gamma g - 2m\gamma v = 0 & c^2 g^2 - m^2 = -1 \\ \gamma^2 (c^2 - v^2) = c^2 & 2c^2 \gamma g = 2m\gamma v & m^2 - c^2 g^2 = 1 \\ \gamma^2 = \frac{c^2}{c^2 - v^2} & c^2 g = mv & m^2 - c^2 g^2 = 1 \\ \gamma = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}} & g = \frac{mv}{c^2} & m^2 - c^2 g^2 = 1 \end{array}$$

The first equation is of no use. Substituting the second equation into the third, we get

$$\begin{aligned} m^2 - c^2 \left(\frac{mv}{c^2} \right)^2 &= 1 \\ \Rightarrow m^2 - \frac{m^2 v^2}{c^2} &= 1 \\ \Rightarrow m^2 \left(1 - \frac{v^2}{c^2} \right) &= 1 \\ \Rightarrow m &= \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}} = \gamma. \end{aligned}$$

We therefore have that $m = \gamma$, and $g = \frac{\gamma v}{c^2}$. We now put these all together.

Theorem 9.1 (Lorentz Transformations). *Suppose that person M is moving at velocity v with respect to person R . Let event E occur at position x_M at time t_M in M 's reference frame. Then, the coordinates of E in R 's reference frame are given by the following equations.*

$$t_R = \frac{\gamma v}{c^2} x_M + \gamma t_M$$

$$x_R = \gamma x_M + \gamma v t_M$$

Alternatively, we have that

$$t_R = \gamma \left(t_M + v x_M \cdot \frac{1}{c^2} \right)$$

$$x_R = \gamma (x_M + v t_M)$$

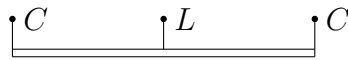
Let us suppose that $v \ll c$. Then, $\gamma \approx 1$. We then get the equations $t_R \approx t_M + \frac{v}{c^2} x_M \approx t_M$, and $x_R \approx x_M + v t_M \approx x_M + v t$. This is the Galilean Transformation.

Remark 9.2. The time dilation and length contraction equations are embedded in the Lorentz Transformations. To see length contraction, suppose that $t_M = t_R = 0$. Then, x_M is the distance person M measures to E , and x_R is the distance that person R measures. Plugging this in to the Lorentz Transformation Equations gives that $x_R = \gamma x_M$. To see time dilation, let $x_M = 0$. t_M is therefore the time that it takes for E to occur in M 's reference frame, and t_R is the time in R 's. Plugging this in to the first equation gives $t_R = \gamma t_M$.

Remark 9.3. The reason that t_R depends on x_M is because of the fact that Leading Clocks Lag. This means that R will see the clock at E lagging, so we will have to add something depending on x_M back in.

9.4 Leading Clocks Lag Revisited

We will let B have a spaceship like the one below.



There are clocks at points marked C , and a launcher at L . Suppose that B is moving at velocity v relative to A , and the launcher shoots light beams in both directions.

9.4.1 Rearward Light Beam

Let t_{AR} be the time that the rearward light beam takes to reach the rear in A 's reference frame, and L_A be the length that she measures for B 's ship. We therefore have that the distance covered is $D = \frac{L_A}{2} - v t_{AR}$. We also have that the elapsed time is $t_{AR} = \frac{D}{c}$. Therefore, we have that

$$c t_{AR} = \frac{L_A}{2} - v t_{AR}$$

$$\Rightarrow t_{AR} = \frac{L_A}{2} \cdot \frac{1}{c + v}.$$

9.4.2 Frontward Light Beam

Let t_{AF} be the time that the rearward light beam takes to reach the read in A 's reference frame, and L_A be the length that she measures for B 's ship. We therefore have that the distance covered is $D = \frac{L_A}{2} + vt_{AF}$. We also have that the elapsed time is $t_{AF} = \frac{D}{c}$. Therefore, we have that

$$\begin{aligned} ct_{AF} &= \frac{L_A}{2} + vt_{AF} \\ \Rightarrow t_{AF} &= \frac{L_A}{2} \cdot \frac{1}{c - v}. \end{aligned}$$

9.4.3 Putting it Together

We have that

$$\Delta t_A = t_{AF} - t_{AR} = \frac{L_A}{2} \cdot \frac{1}{c - v} - \frac{L_A}{2} \cdot \frac{1}{c + v} = \frac{L_A v}{c^2 - v^2}.$$

We can rearrange this to get that

$$\begin{aligned} \Delta t_A &= \frac{L_A v}{c^2} \gamma^2 \\ \Rightarrow \frac{1}{\gamma} \Delta t_A &= \frac{(\gamma L_A) v}{c^2} \\ \Rightarrow \Delta t_B &= \frac{L_B v}{c^2}. \end{aligned}$$

This gives the following theorem.

Theorem 9.4. *If B is moving with velocity v with respect to A , and B has 2 clocks that he measures to be at a distance L_B apart, then A measures the lag between the two clocks to be $\frac{L_B v}{c^2}$.*

9.4.4 Proof with Lorentz Transformation

Let us suppose that there is a clock reading t_B in A 's reference frame at position x_B (in B 's reference frame) at $t_A = 0$. We wish to find t_B . This is equal to $\gamma(t_A + \frac{v}{c^2}x_A)$. Because $t_A = 0$, we have that this is equal to $\gamma \cdot \frac{v}{c^2}x_B = \frac{vx_B}{c^2}$. However, $x_B = L_B$, so $\Delta t_B = \frac{L_B v}{c^2}$.

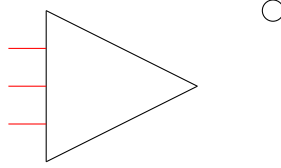
9.5 Time Dilation and Length Contraction Example (Part 2)

We saw that, when the rocket reaches, they read Earth's clocks as saying 0.59. However, when they reach, the star clock has to read 5.3. This is answered by Leading Clocks Lag. The rocket will observe the lag as $\frac{5 \text{ lightyears} \cdot 0.943c}{c^2} = 4.71$ years. Now, if we add the lag to the time on Earth's clock, we get $4.71 + 0.59 = 5.3$. This means that the star clock reads 5.3, and so there are no inconsistencies.

10 Combining Velocities

10.1 Parallel Velocities

Let us suppose that we have a rocket moving at velocity v (with respect to the L frame) that launches an object at velocity u_R relative to the rocket. We define x_R to be the position of the launched object with respect to the rocket, and t_R to be the time (in the rocket's reference frame) that the object is at x_R . We also define x_L to be the position of the object in the L reference frame, and t_L the time at which that happens.



We have that $u_R = \frac{\Delta x_R}{\Delta t_R}$. We want to find $u_L = \frac{\Delta x_L}{\Delta t_L}$. Lorentz Transformations give that $\Delta x_L = \gamma(\Delta x_R + v\Delta t_R)$ and that $\Delta t_L = \gamma(\Delta t_R + \frac{v}{c^2}\Delta x_R)$. Therefore,

$$u_L = \frac{\gamma(\Delta x_R + v\Delta t_R)}{\gamma(\Delta t_R + \frac{v}{c^2}\Delta x_R)} = \frac{\Delta t_R \left(\frac{\Delta x_R}{\Delta t_R} + v \right)}{\Delta t_R \left(1 + \frac{v}{c^2} \frac{\Delta x_R}{\Delta t_R} \right)} = \frac{u_R + v}{1 + \frac{u_R v}{c^2}}.$$

This is the relativistic velocity addition formula.

Theorem 10.1 (Relativistic Addition of Velocity). *Let us suppose that D is moving with velocity u with respect to B , and that B is moving with velocity v with respect to A . Then, the velocity of D with respect to A is*

$$\frac{u + v}{1 + \frac{uv}{c^2}}.$$

Remark 10.2. This formula only applies in 1 dimension.

Remark 10.3. If $u, v \ll c$, then the formula is $\approx u + v$.

Remark 10.4. If $u = c$, then the formula is

$$\frac{c + v}{1 + \frac{v}{c}} = \frac{c + v}{c + v} \cdot c = c.$$

10.2 Perpendicular Velocities

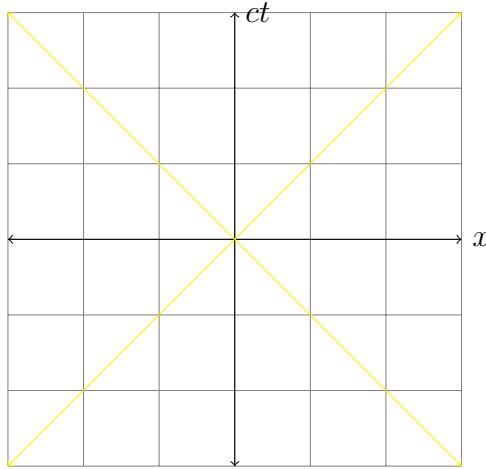
Let us suppose that u_R is perpendicular to v . We then have that $\Delta u_{RY} = \frac{\Delta y_R}{\Delta t_R}$, $u_{RX} = \frac{\Delta x_R}{\Delta t_R} = 0$, and $\Delta u_{LY} = \frac{\Delta y_L}{\Delta t_L}$. Because there is no width contraction, we also have that $\Delta y_R = \Delta y_L$. Lorentz transformations give that $\Delta t_L = \gamma \left(\Delta t_R + \frac{v}{c^2} \cdot \Delta x_R \right)$. Thus, we have that

$$u_{LY} = \frac{\Delta y_L}{\Delta t_L} = \frac{\Delta y_R}{\gamma \left(\Delta t_R + \frac{v}{c^2} \cdot \Delta x_R \right)} = \frac{\Delta y_R}{\gamma \Delta t_R} = \frac{\Delta u_{RY}}{\gamma}.$$

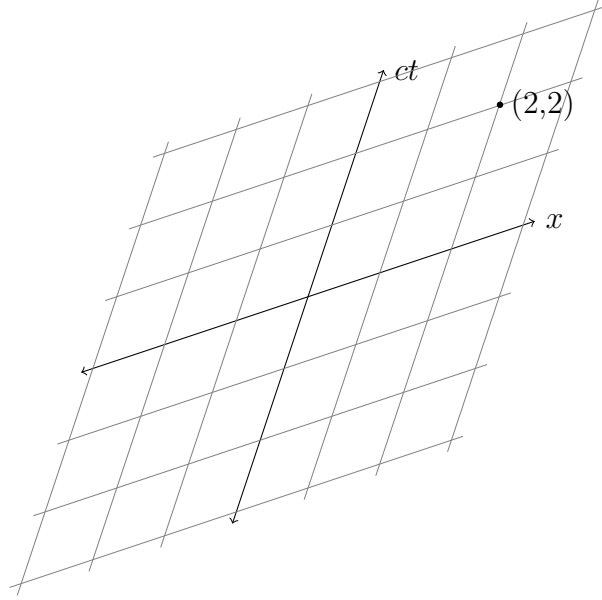
We also have that $u_{LX} = \frac{0+v}{1+0 \cdot \frac{v}{c^2}} = v$.

11 Spacetime Diagrams

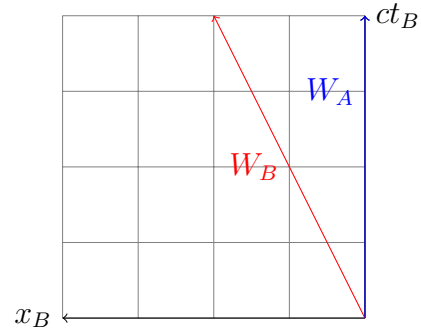
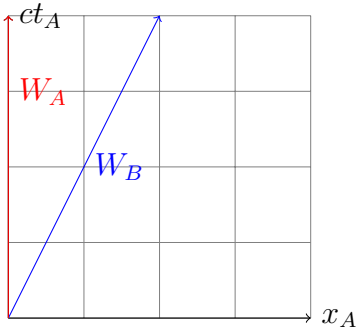
Note that in a spacetime diagram, lines of same location are $x = C$ (vertical), and lines of same time are $t = C$ (horizontal). We can also see that, if 2 worldlines intersect, then the objects are in the same spot at the same time. Recall the units in which $c = 1$ (which we will use these units for the rest of the section). Those give the spacetime diagram below.



We also have modified coordinates as below.

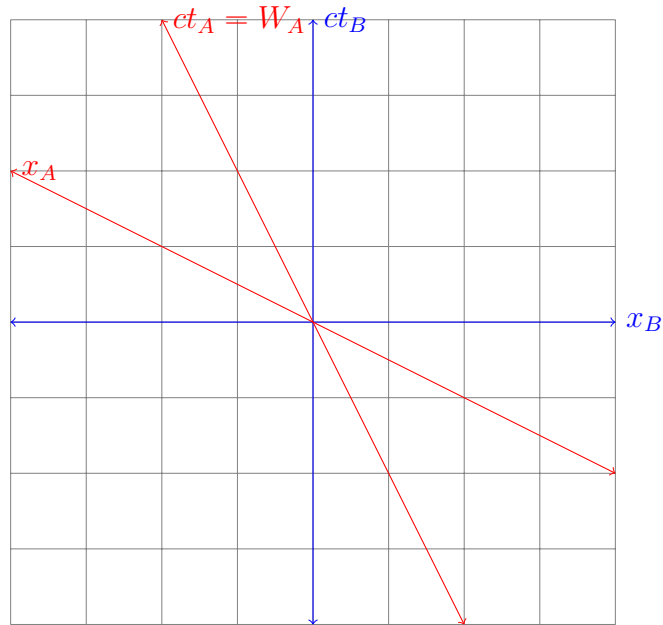
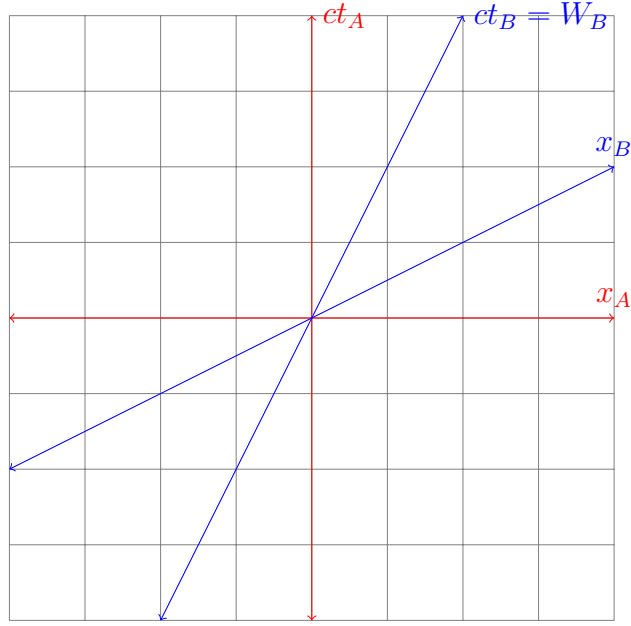


Recall how we change reference frames.



11.1 Transformations

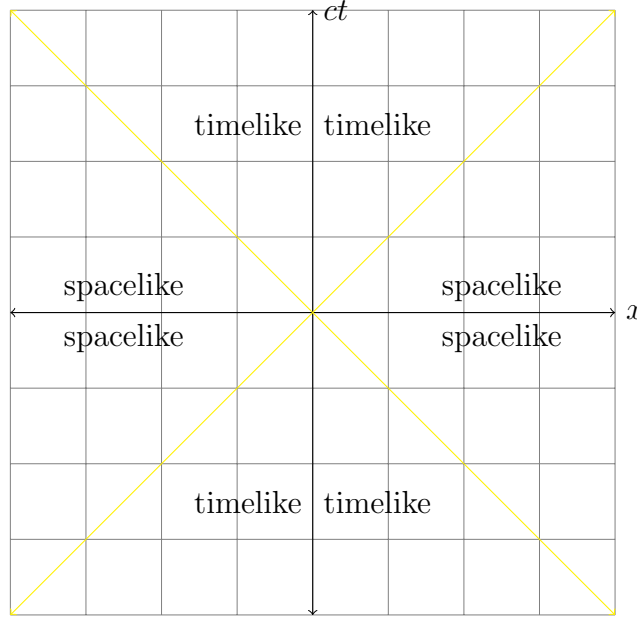
We will use Lorentz Transformations to visualize coordinate transformations. Suppose that B is moving at velocity v with respect to A . The t_B -axis is the line $x_B = 0$. Lorentz Transformations give $\gamma(x_A - vt_A) = 0$, or $t_A = \frac{1}{v}x_A$. Meanwhile the x_B -axis is the line $t_B = \gamma(t_A - \frac{v}{c^2}x_A) = 0$, which is the line $t_A = \frac{v}{c^2}x_A$. We can see the transformation below.



These allow us to see why simultaneity is relative. Lines of simultaneity in B 's reference frame are parallel to x_B , and lines of simultaneity in A 's reference frame are parallel to x_A . However, x_A and x_B are not parallel, so simultaneity is relative.

11.2 Intervals

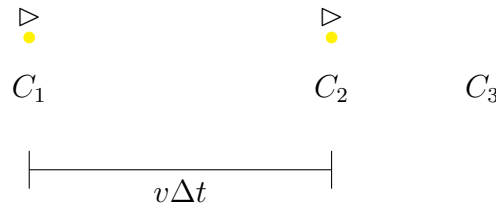
We define the interval between (x_0, t_0) and (x_1, t_1) to be *timelike* if $c^2(t_1 - t_0)^2 - (x_1 - x_0)^2 > 0$, *spacelike* if $c^2(t_1 - t_0)^2 - (x_1 - x_0)^2 < 0$, and *lightlike* if $c^2(t_1 - t_0)^2 - (x_1 - x_0)^2 = 0$.



12 Faster than Light Travel

12.1 A Possible Way

Let us suppose that a rocket moves from city C_1 to C_2 at velocity v . When it is at both points, it sends a light pulse to C_3 . We can see the diagram below.



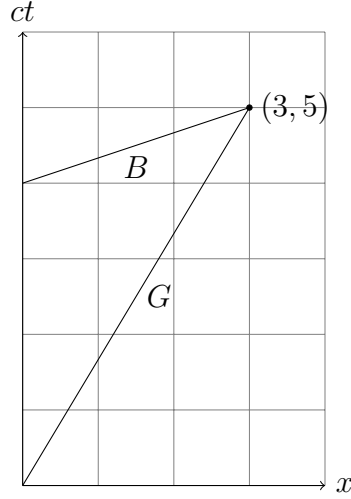
The distance between the light pulses is $c\Delta t - v\Delta t = (c - v)\Delta t$. Therefore, C_3 will measure the time difference between the flashes to be $\frac{(c-v)\Delta t}{c}$. This means that C_3 will measure the speed of the rocket to be $v\Delta t \cdot \frac{c}{(c-v)\Delta t} = \frac{cv}{c-v}$. If we use the units in which $c = 1$, we get that this is equal to $\frac{v}{1-v}$. If we let $v = 0.8c$, then this is equal to $4c$. However, this is just an optical illusion.

12.2 Breaking of Causality

Let us suppose that the following sequence of events occur:

- (1) At $t_B = 0$, the good guys and the bad guys sign a peace treaty, and the good guys leave at $v = 0.6c$.
- (2) At $t_B = 4$, the bad guys develop a faster-than-light spaceship, and go after the good guys at $v = 3c$.

(3) At $t_B = 5$, the bad guys catch up and launch a sneak attack.

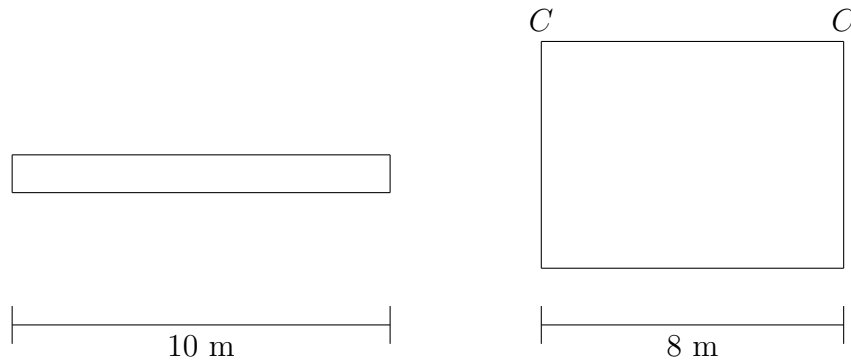


Let us find when the good guys observe the sneak attack. They see it at $x_G = \gamma(x_B - vt_B) = 1.25(3 - 0.6 \cdot 5) = 0$, and at $t_G = \gamma(t_B - \frac{v}{c^2}x_B) = 1.25(5 - 0.6 \cdot 3) = 4$. Now, let us find when the good guys observe the creation of the spaceship. They will see this at $t_G = \gamma(t_B - \frac{v}{c^2}x_B) = 1.25(4 - 0) = 5$. This means that the good guys will see the spaceship reaching them before it was even created, which breaks causality.

13 Paradoxes

13.1 Pole in the Barn Paradox

We set up a situation as below.



We will have the pole move at speed $v = 0.6c$ towards the barn, so that $\gamma = 1.25$. The barn will measure the length of the pole to be $\frac{10\text{m}}{1.25} = 8\text{ m}$. Thus, when the barn observes the pole to fully be inside it, it will briefly close the doors. However, to the pole, it will never fit inside the barn fully, so this creates an apparent paradox. We assume that $t_A = t_B = 0$ when the front of the pole is just entering the barn.

13.1.1 The Barn's Reference Frame

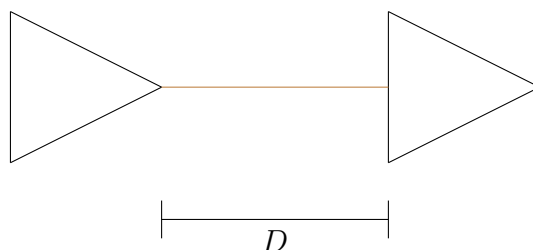
The barn will see the pole take $\frac{8m}{0.6c} = 44.4$ ns for the front of the pole to reach the back door (or the back of the pole to reach the front door).

13.1.2 The Pole's Reference Frame

The pole will see the length of the barn as $\frac{8}{\gamma} = 6.4$ m. It will therefore see the back door take $\frac{6.4 \text{ m}}{0.6c} = 35.6$ ns to reach it, and will see the barn's clock tick off $\frac{35.6 \text{ ns}}{\gamma} = 28.4$ ns. However, the pole will see the clock at the rear door lag by $\frac{8 \text{ m} \cdot 0.6c}{c^2} = 16$ ns. Therefore, when the pole reaches the rear door, the clock there will read 44.4 ns. Now, the pole sees the front door having to travel another 3.6 m, and will see it take another $\frac{3.6}{0.6c} = 20$ ns. Therefore, the front clock will read $\frac{35.6+20}{\gamma} \text{ ns} = 44.4$ ns.

13.2 Spaceships on a Rope Paradox

Let us suppose that we have a setup similar to the one below.



We will have A outside, B in the spaceship on the left, and C in the spaceship on the right, with a taught rope between B and C . On command from A , we will have B and C accelerate so that the distance between them stays D in A 's reference frame. We want to figure out if the rope breaks.

13.2.1 The Rockets' Reference Frame

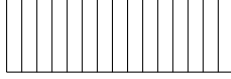
Let us suppose that B and C accelerate at the same time. Then, because leading clocks lag, A will observe C accelerate after B . Therefore, C has to accelerate before B in the rockets' reference frame. This will snap the rope, creating an apparent paradox.

13.2.2 A 's Reference Frame

A will see the distance between B and C stay the same. However, she will also see the rope shrink because of length contraction, which will cause the rope to snap.

13.3 How Does Length Contraction Happen?

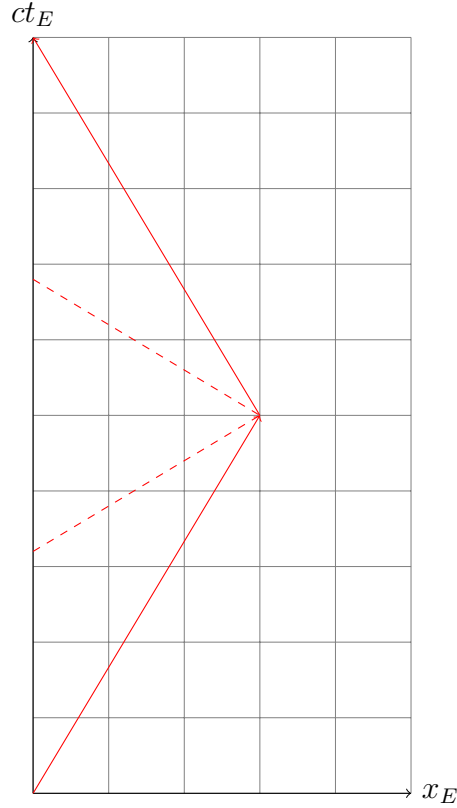
Let us suppose that we have a rod traveling at velocity v , and it accelerates to velocity $v + \Delta v$. We want to figure out how length contraction happens. To see this, divide up the rod as below.



If the pieces accelerate simultaneously relative to the rod, then a stationary³ observer will see the pieces in the back accelerate first. This is what causes contraction.

13.4 The Twin Paradox

Let us suppose that A leaves B who is on Earth in a spaceship at $v = 0.6c$ to a star 3 lightyears away, and then returns, as in the diagram below.



13.4.1 B 's Reference Frame

B sees A take $\frac{3 \text{ lightyears}}{0.6c} = 5$ years, and observes her clock tick off $\frac{5 \text{ years}}{\gamma} = 4$ years. This is true on both the departure and return journeys.

13.4.2 A 's Reference Frame

A sees the star travel $\frac{3 \text{ lightyears}}{\gamma} = 2.4$ lightyears, and sees it take $\frac{2.4 \text{ lightyears}}{0.6c} = 4$ years. She also sees B 's clock tick off $\frac{4 \text{ years}}{\gamma} = 3.2$ years on both trips.

³moving with velocity $-v$ in the rod's RF

13.4.3 Paradox and Resolution

We previously resolved this using the fact that leading clocks lag. However, because A returns to B , this argument doesn't work. The key is that A turns around, which means that she is not on the same footing as B . During the turning, B 's clocks rapidly increase (the portion between the dashed lines). The reason for this rapid increase is because, during the turn, the leading clock turns from the one on Earth to the one at the star, so the Earth clock must rapidly increase (by 2 times the lag). Computing 2 times the lag gives $2 \cdot \frac{3 \cdot 0.6c \text{ lightyears}}{c^2} = 2 \cdot 1.8 \text{ years} = 3.6 \text{ years}$. Sure enough, $3.2 + 3.2 + 3.6 = 10$, so there is no paradox.

13.5 General Solutions

We have seen the answer to the Pole in the Barn and Twin Paradoxes for special values. We will now use variables, and show that there was not any manipulation.

13.5.1 Pole in the Barn

Let us suppose that, at rest, the pole has length L . Then, the barn must have length $\frac{L}{\gamma}$ in order for the paradox to apply.

Back Door The barn will observe the time at which the back door shuts to be $\frac{L}{\gamma v}$. The pole will observe the time to be $\frac{L}{\gamma^2} \cdot \frac{1}{v}$. The pole will therefore see the barn's clock tick off $\frac{1}{\gamma} \cdot \frac{L}{\gamma^2 v}$. However, the pole will observe a lag of $\frac{Lv}{\gamma c^2}$, and will thus see the clock at the back door read

$$\frac{L}{\gamma^3 v} + \frac{Lv}{\gamma c^2} = \frac{L}{\gamma v} \left(\frac{1}{\gamma^2} + \frac{v^2}{c^2} \right) = \frac{L}{\gamma v} \left(1 - \frac{v^2}{c^2} + \frac{v^2}{c^2} \right) = \frac{L}{\gamma v}$$

when it shuts. There is therefore no paradox here.

Front Door The barn will observe the time at which the front door shuts to be $\frac{L}{\gamma v}$. The pole will observe it to be $\frac{L}{v}$, and will therefore see the barn's clock tick off $\frac{L}{\gamma v}$, so there is no paradox here either.

13.5.2 Twin Paradox

We will suppose that the distance to the star is x .

B 's RF B sees the entire trip taking $\frac{2x}{v}$, and sees A 's clocks ticking off $\frac{2x}{\gamma v}$.

A 's RF A sees the entire trip taking $2 \cdot \gamma \left(\frac{x}{v} - \frac{v}{c^2} x \right)$ (by the Lorentz Transformation). This is equal to $\frac{2\gamma x}{v} \left(1 - \frac{v^2}{c^2} \right) = \frac{2x}{v} \cdot \gamma \cdot \gamma^{-2} = \frac{2x}{\gamma v}$. This means that A will see B 's clocks ticking off $\frac{2x}{v\gamma^2}$. We must add in the lag, which is $\frac{2xv}{c^2}$, so we get that the elapsed time is $\frac{2x}{v\gamma^2} + \frac{2xv}{c^2} = \frac{2x}{v} \left(\frac{1}{\gamma^2} + \frac{v^2}{c^2} \right) = \frac{2x}{v} \left(1 - \frac{v^2}{c^2} + \frac{v^2}{c^2} \right) = \frac{2x}{v}$, so there is no paradox.